

Theory of Computation

Submission link: <https://forms.gle/9QfcWigewGmCcvG36>

Exercise 11_12: (Pushdown Automata & Properties of CFL)

1. Find the language of NPDA M.

$$L = \{0\} \cup \{011^*0\}$$

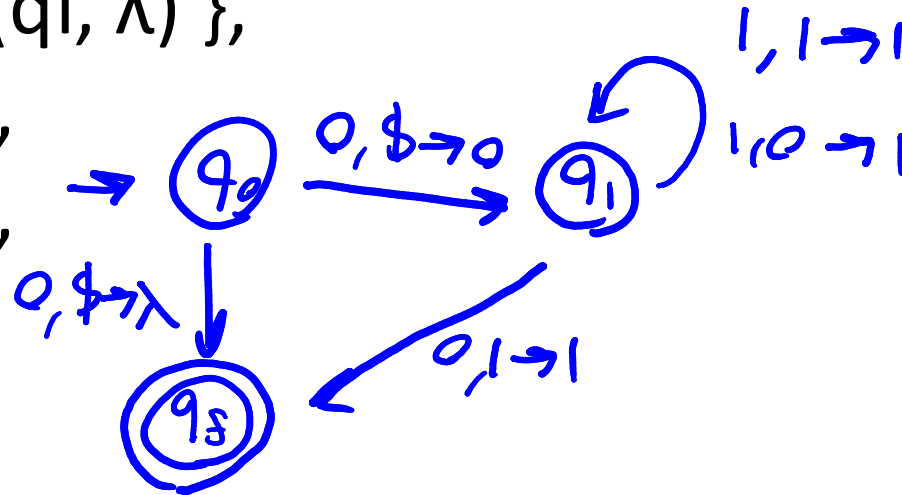
$$M = (\{q_0, q_1, q_f\}, \{0, 1\}, \{0, 1, \$\}, \delta, q_0, \$, \{q_f\})$$

$$\delta: \delta(q_0, 0, \$) = \{(q_1, 0), (q_f, \lambda)\},$$

$$\delta(q_1, 1, 1) = \{(q_1, 1)\},$$

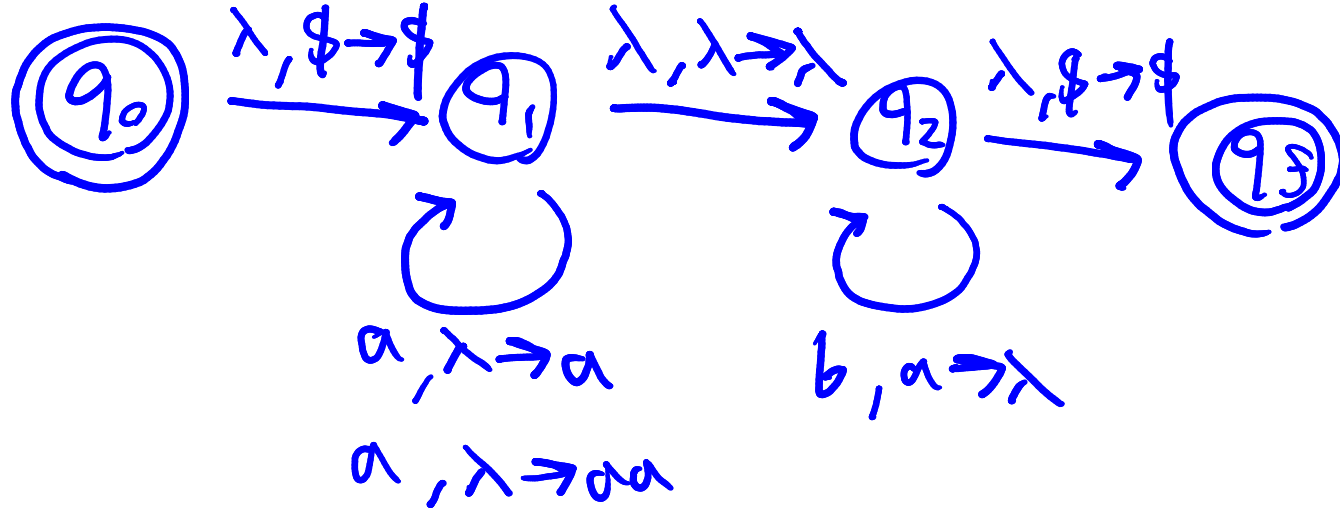
$$\delta(q_1, 1, 0) = \{(q_1, 1)\},$$

$$\delta(q_1, 0, 1) = \{(q_f, 1)\}$$



2. Find PDA for the language L2.

$$L2 = \{a^n b^m : n \leq m \leq 2n\}$$



*3. Prove that the language L is CFL by using properties of CFL (DO NOT draw PDA or CFG).

$$L3 = \{0^i 1^j 2^k : j = i + k\}$$

$$\underbrace{0^i 1^i}_{\text{CFL}} \underbrace{1^k 2^k}_{\text{CFL}}$$

Concat of CFL $\rightarrow$ Is CFL!